

Unlocking New Opportunities through Location Insights



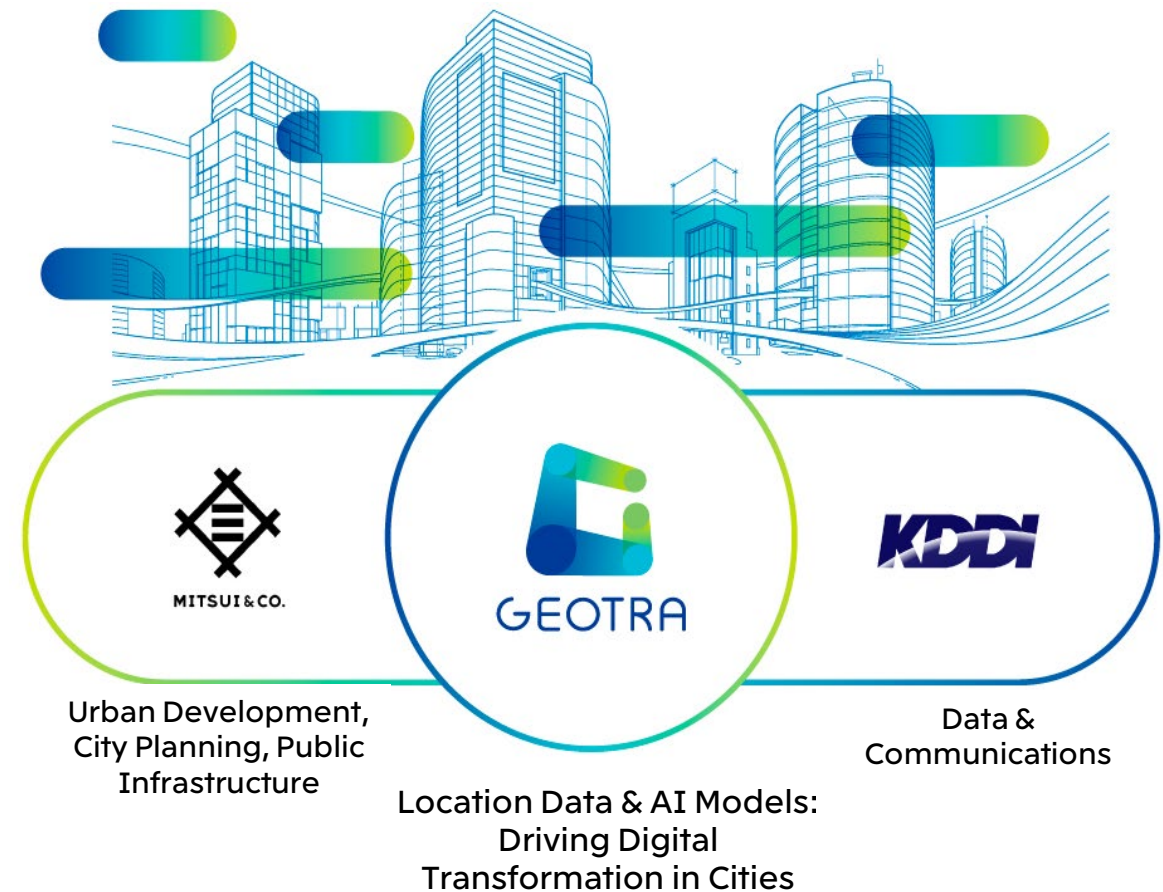
Mitsui & Co., Ltd.

Integrated Energy Solutions Business Unit

Driving society forward through the power of data

With deep expertise in geolocation big data, we support our clients in solving complex business challenges and unlock new opportunities

GEOTRA is a joint venture between Mitsui & Co., a global leader in diverse business sectors, and KDDI, a powerhouse in big data and digital transformation. We drive forward by leveraging the extensive networks and assets of our parenting companies.



At GEOTRA, we combine diverse location data sets to address client challenges.

Human Mobility Data



Vehicle Data



Payment Data



Logistics Data



IoT Data



Map Data



As experts in leveraging location intelligence, GEOTRA provides data-driven solutions that address complex business challenges

Data

Human Mobility Data

Deep Insights into Human and Social Dynamics



- GEOTRA Activity Data
- KDDI Location Analyzer
- Tourists Data

Consulting

Guidance Every Step of the Way : Identifying challenges~ data selection ~ analysis and implementation



- Payment / Consumer spending data
- Transportation (vehicle, bus, railway) Data
- In-store Customer Flow Data
- Human mobility data

AI / Engineering

AI Simulation

AI powered Mobility Forecasting



- GEOLABO <Urban Planning Simulation>
- Simulation on disaster prevention, transportation model

Infrastructure Development & Digital Transformation Solutions

Supporting Business Infrastructure Development



- Custom digital solutions & development services
- Dashboard development for data visualization
- Integration with 3D urban models

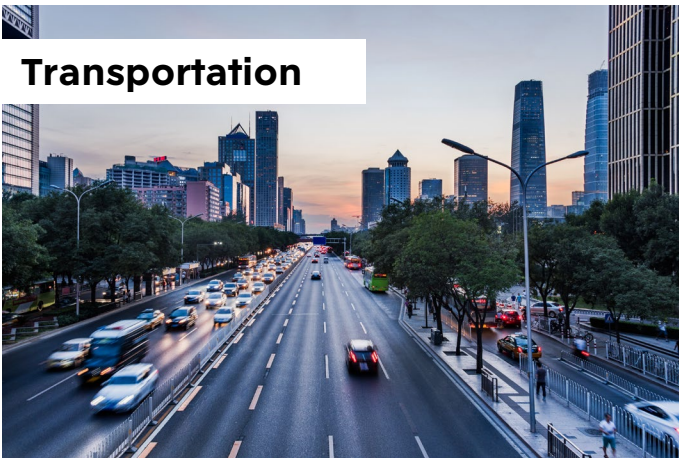
Business Overview

Urban planning



Analyzing the excursion of cities and the individual traits of their visitors can provide evidence to promote urban redevelopments.

Transportation



Visualizing the usage of public transportation and utilizing the data to relieve traffic.

Construction



Understanding the specific usage of urban infrastructure such as time-worn bridges and roads to conduct effective maintenance.

Tourism



Utilizing the trails of tourists to introduce future strategies. Cross-analysis with inbound data can also be effective.

Disaster Prevention



Simulating people's activities in cases of emergency. We can utilize the data when determining the priority of roads to operate.

Marketing



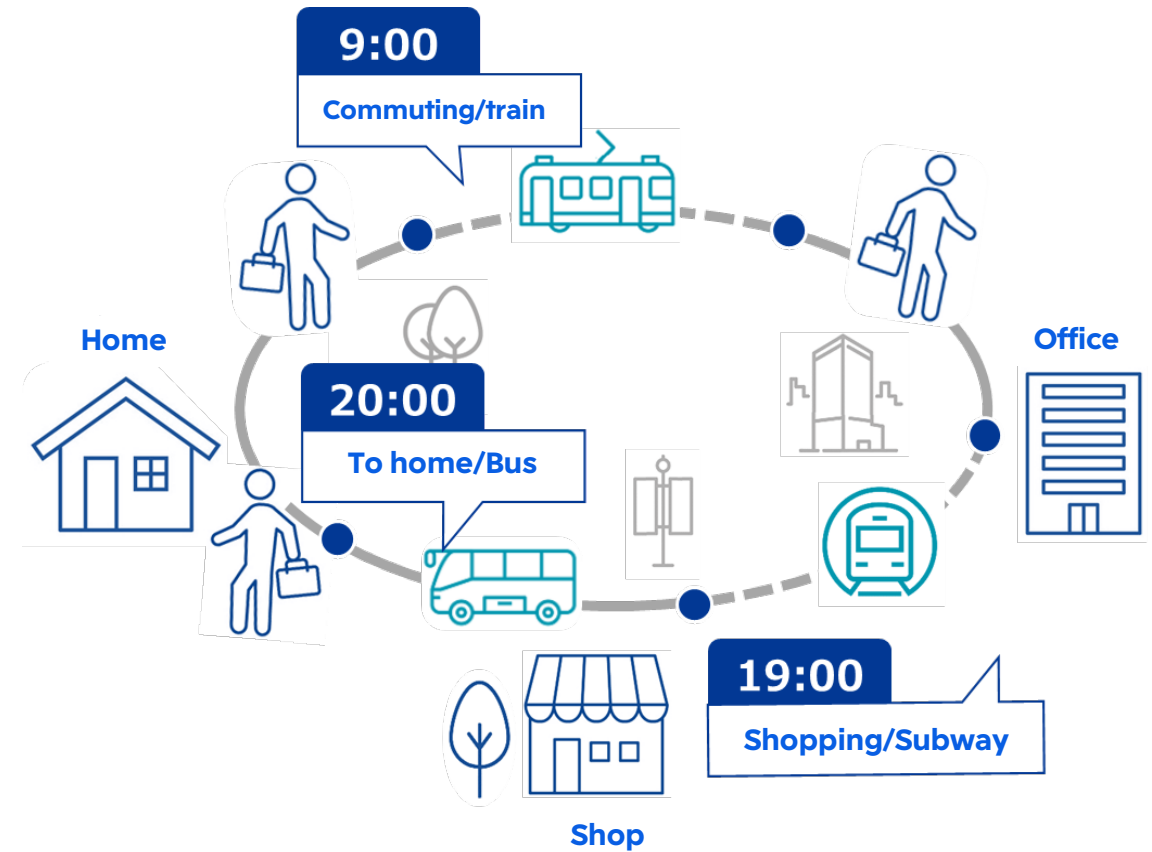
Measuring the impact of certain events to promote marketing.

Human Mobility Data



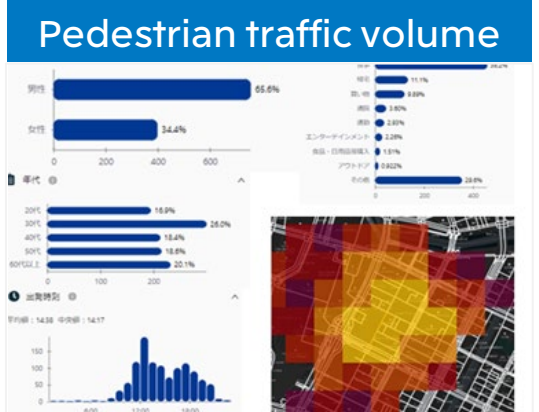
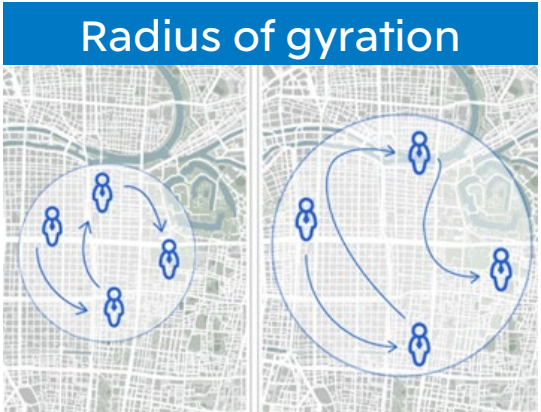
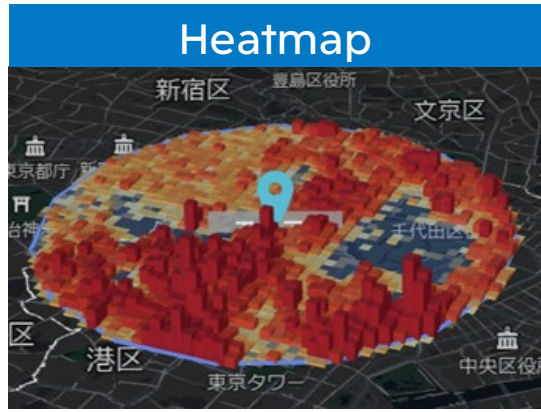
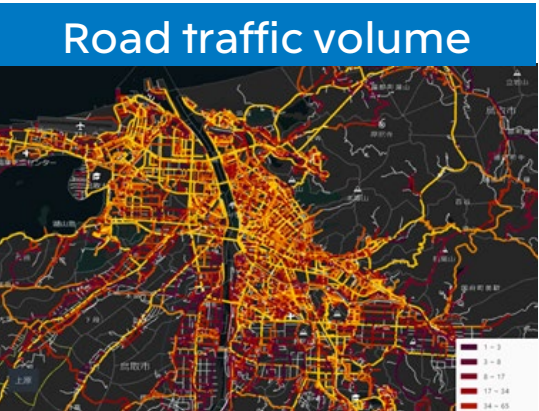
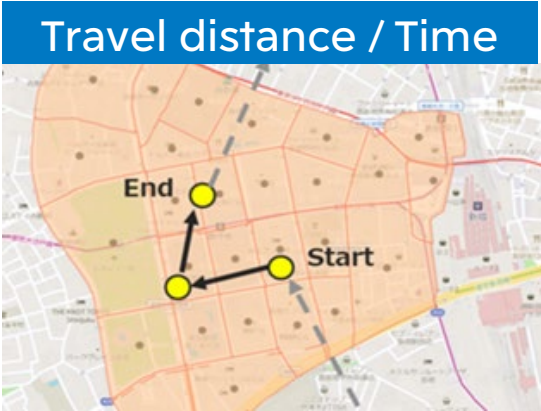
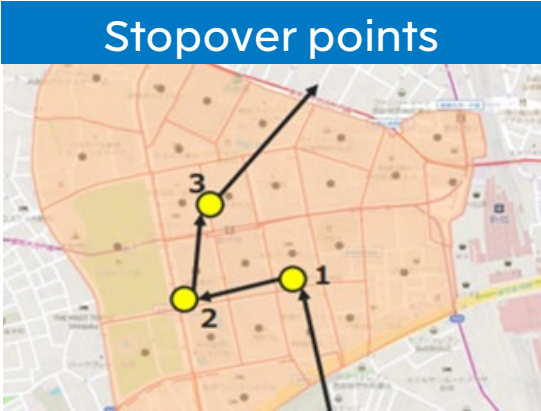
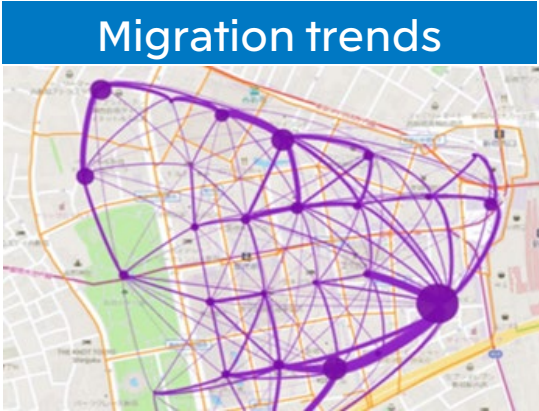
“Human Mobility Data” that reproduces individual movements/flows at a high granularity

GEOTRA provides human mobility data and develops insight-driven services based on location intelligence.



Human Mobility Data Products – GEOTRA Activity Data Analysis Examples

We are able to deliver valuable insights by analyzing and aggregating human mobility data from multiple perspectives.



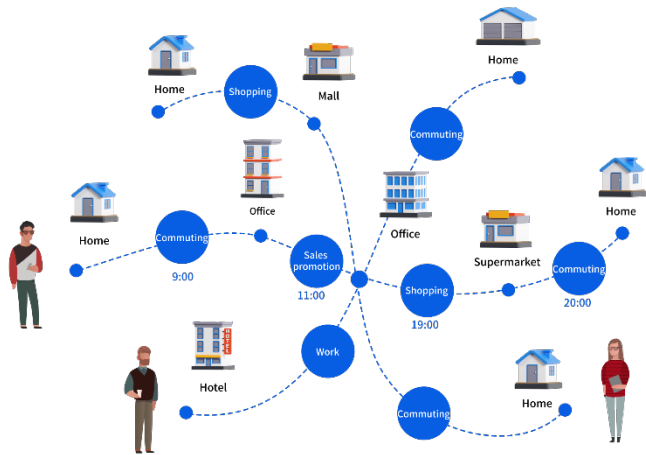
※ Some analysis examples are not standard features on the web dashboard.
We may propose a consulting plan after discussing your objectives and background.

GEOTRA's Data Overview

GEOTRA's data overcomes the typical challenges of analyzing activities by enabling flexible analysis and simulations, while simultaneously protecting individuals' privacy.

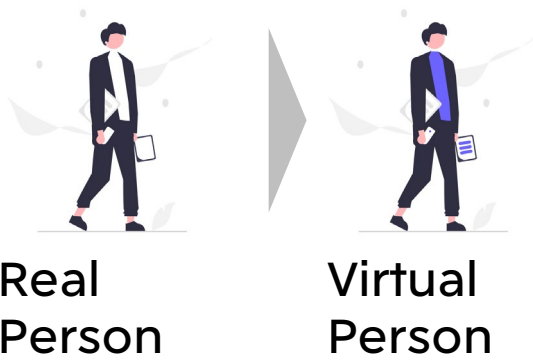
Flexible analysis (Disaggregated trip data)

GEOTRA's data can visualize the flow of people within the area instead of only the number of people.



Protects individuals' privacy (Virtual Person)

Synthetic data generates virtual person which show the same traits as people in real life



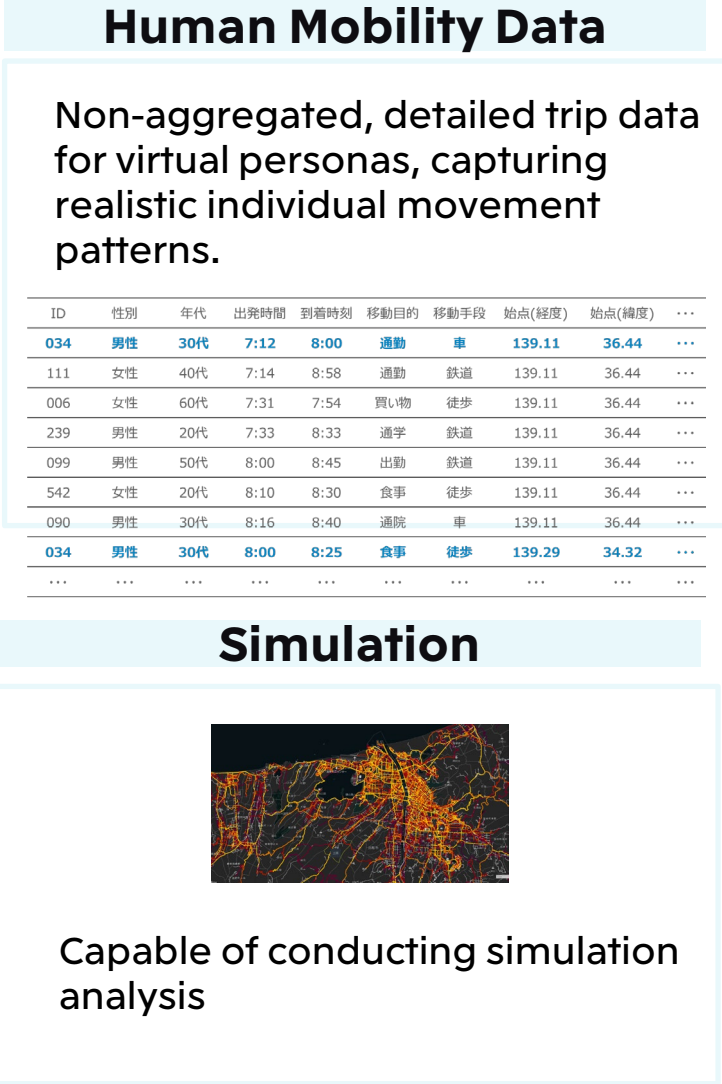
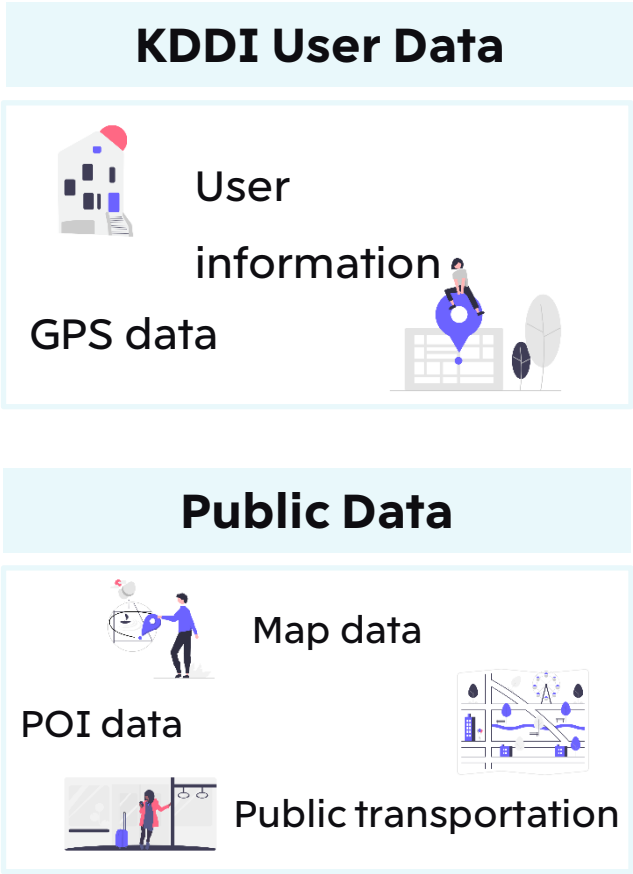
Simulates activities of individuals (AI modeling)

With AI modeling technology,
we simulate activities of virtual
persons – we forecast and
display how their activities
change under certain conditions



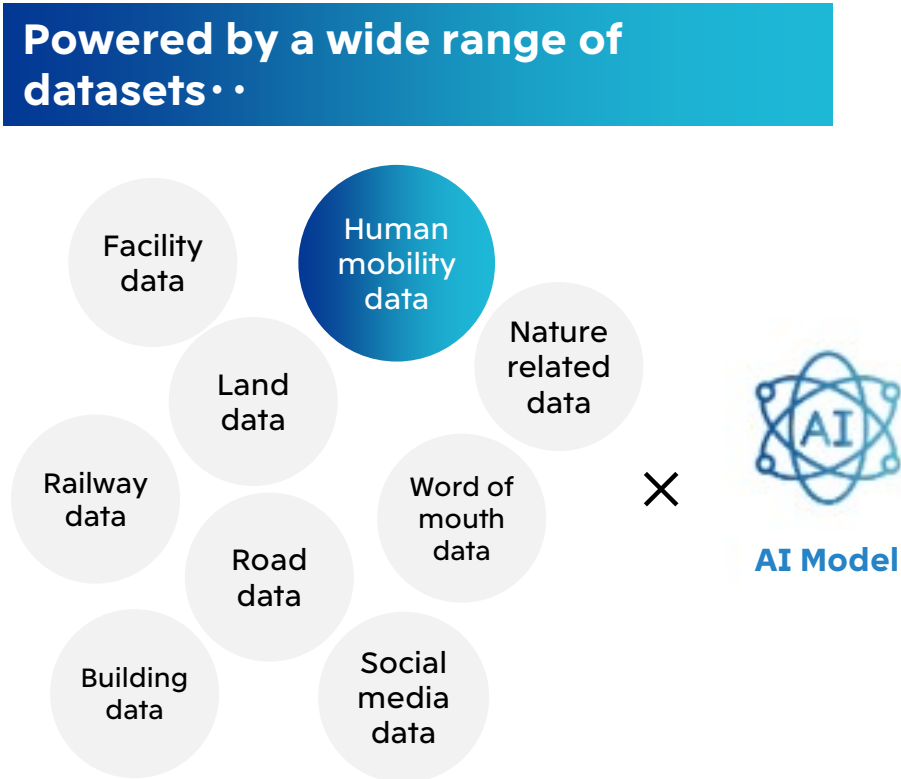
GEOTRA's Data Overview

We combine KDDI location data and various public datasets with advanced data processing and privacy protection technologies to generate individual movement data.



AI Simulation

Using a variety of geospatial and human mobility data, combined with AI, we predict how people's movements will change in response to urban developments.



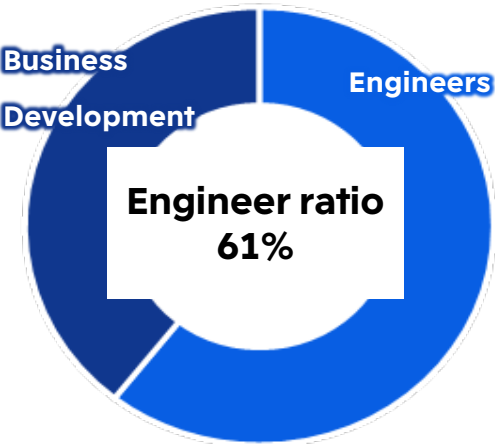
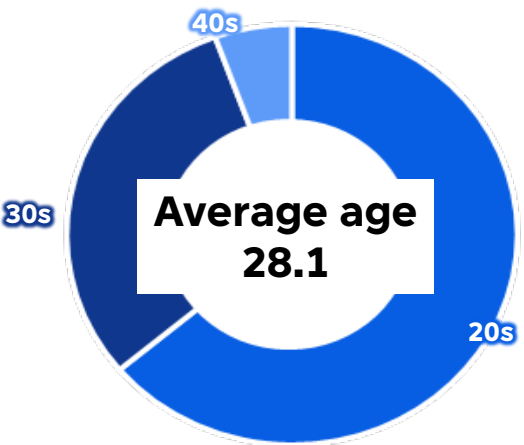
Predicting people's movement



Technical Capability

Our team of consultants and data scientists, with deep expertise in mathematical modeling, provides robust technical solutions grounded in a wealth of project experience.

Organizational Demographics



Technical capabilities

Statistical model	Machine Learning	Mathematical Optimization	Network Analysis
•Regression Analysis •Time Series Analysis •Spatial Point Analysis •Markov Model •Spatial Statistics •Geographically Weighted Regression (GWR)	•Decision Tree •Random Forest •Gradient Boosting •LightGBM •SVM •Clustering	•Meta-Heuristics •Simulated Annealing	•Road NW Analysis •Transportation NW Analysis
		Agent Simulation	Privacy Protection Technology
		•Multi-Agent Simulation •Activity-based model	•Synthetic Data •k-Anonymity

Achievement

Our proven track record includes lots of academic research collaborations and frequent presentations at leading conferences.

AI・データサイエンス論文集
2023年4巻3号1-9

GeoNews 2023.05.17

ASCII 2023.7.31

日経XTECH 2024.08.02

We support a wide range of clients, from leading private companies to public sectors.

Our Clients (Examples)

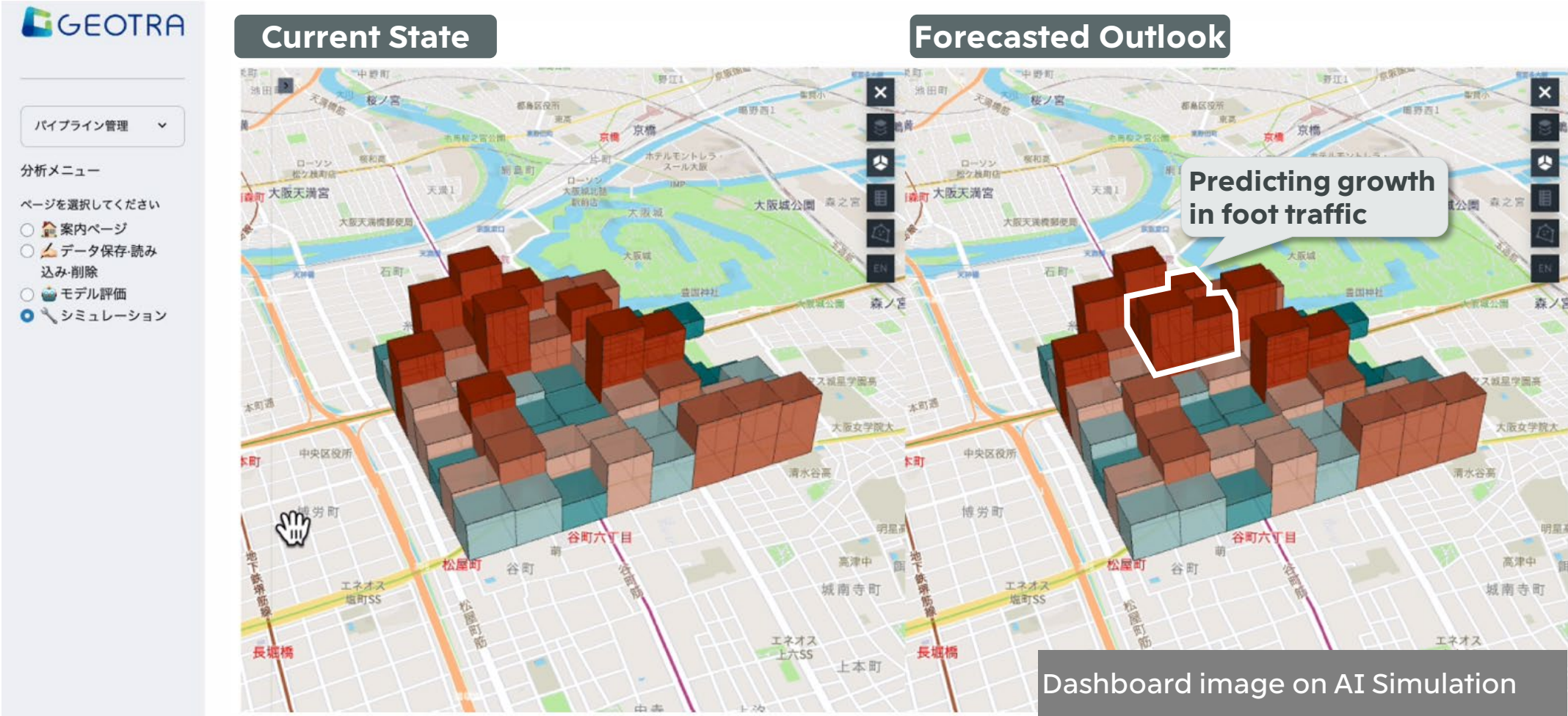


Category	Model Description	Customers
Urban Redevelopment	- Simulating changes in foot traffic after major central city redevelopment projects - Modeling the impact on foot traffic and vibrancy after creating new gathering spaces	Construction consulting firms Real estate developers, government agencies
Facility & Land Use Planning	Simulating human mobility changes from the opening of large scale hospital	Consulting firms, government agencies
Mobility Shifts	Simulation for optimizing the placement and forecasting the impact of modal shift strategies toward public transportation	Railway operators
Road Infrastructure Renewal	Simulating traffic impacts from major road renovations and lane closures	Road infrastructure companies, research institutions
Congestion Mitigation	Forecasting the impact of congestion mitigation measures and new road construction	Government agencies
Disaster & Emergency Planning	- Modeling evacuation patterns during large earthquakes - Simulating road clearance operations after severe flood damage	Construction companies, research institutions

Sample case

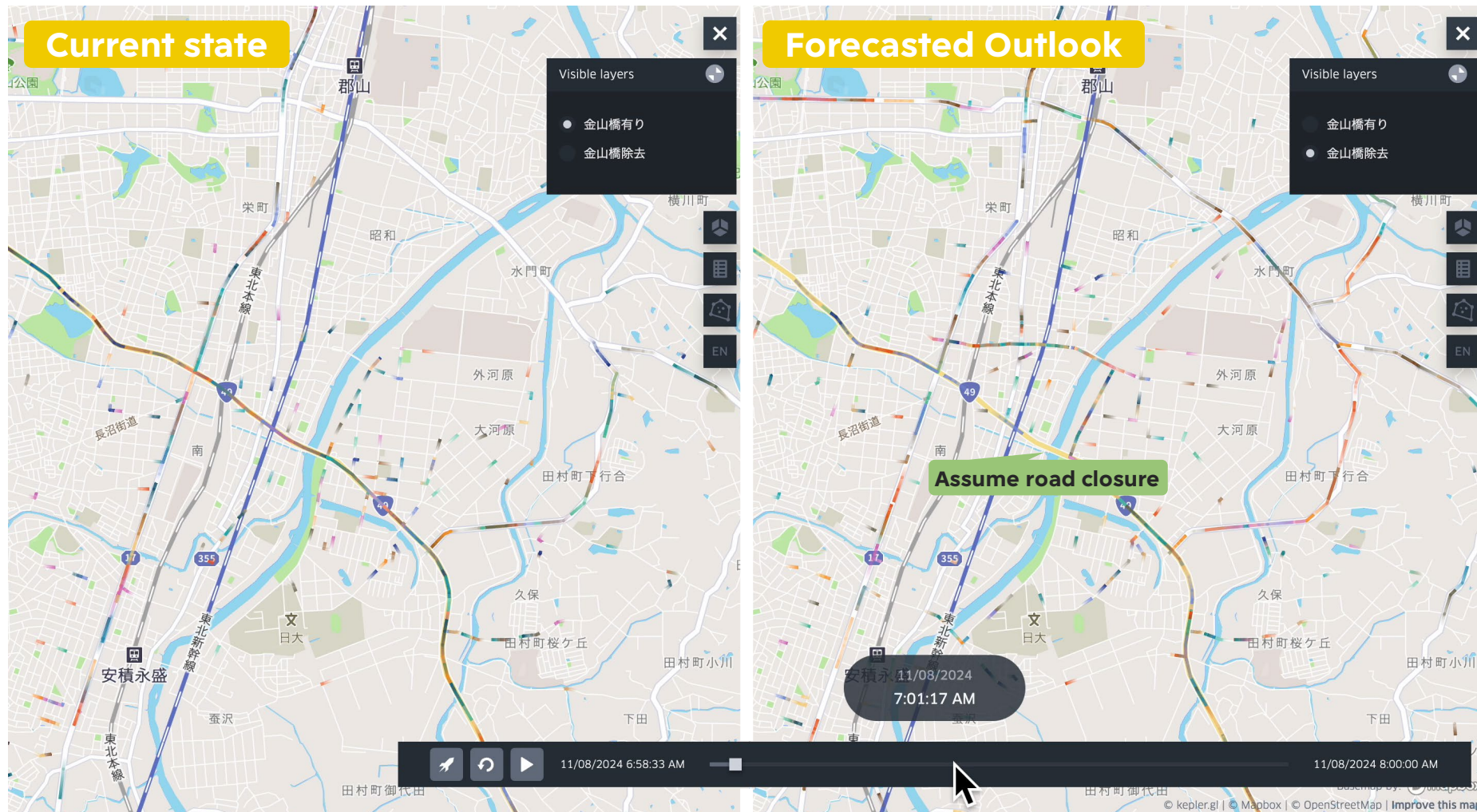
Understanding Redevelopment Impact : Before vs After

We help predict how foot traffic will change before and after redevelopment – enabling to assess the impact early and better to understand the local area.



Making Smarter Infrastructure Decisions

We support data-driven decisions on which assets matter most and how to maintain them by forecasting traffic changes and potential economic loss during infrastructure updates (e.g., road closures).



Infrastructure Development & Digital Transformation Solutions

Infrastructure Development & DX Support

Based on insights derived from our location intelligence and big data solutions, we assist clients in effectively utilizing their data assets.



Example of Human Flow Simulation
Built on a 3D Urban Model

Technical Stack

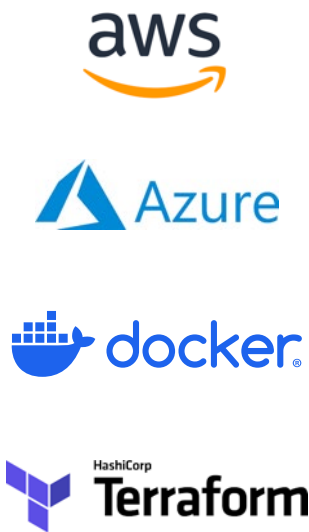
Frontend



Backend



Infrastructure



Analytics

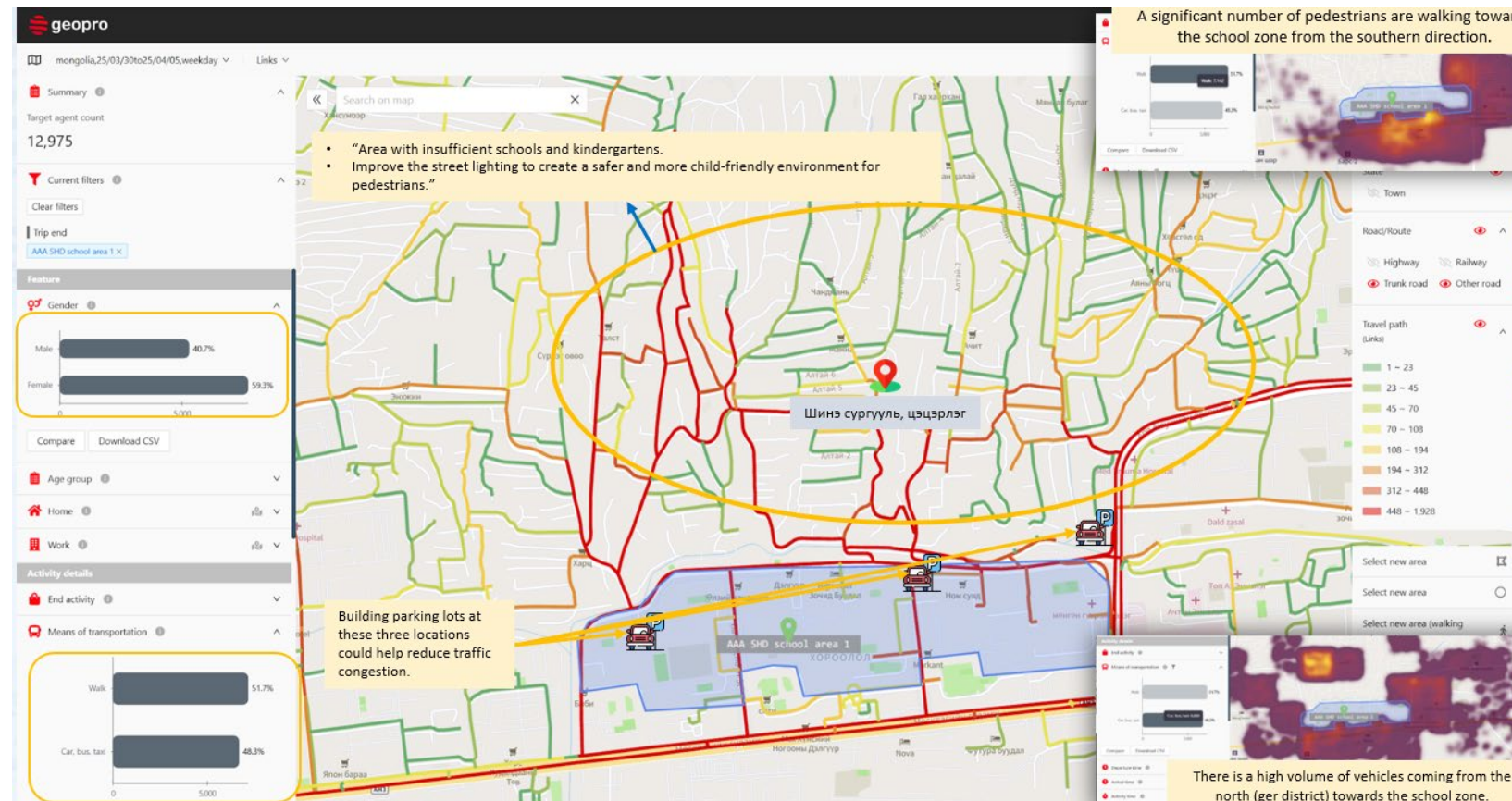


Supporting Daily Operations with a Robust IT Foundation

	Issues		What We Can Provide	Tool Examples
Data Utilization Strategy	Unclear how data is used internally	Data Mapping	Analyze existing reports/tools to map who uses what data, when, and for what purpose	Consulting
	No clear use cases for data-driven decisions	Use Case Exploration	Explore diverse data use cases based on available data and customer business insights	Consulting
	Hard to prioritize use cases	Use Case Assessment	Evaluate each use case from multiple angles to identify top priorities	Consulting
	Database structure needed to support each use case is unclear	Data Modeling	Design and build optimal data models aligned with prioritized use cases	dbt
Building Data Infrastructure	Improve data use without system overhaul	Data Replication	Build a secure system to replicate and sync data from existing systems (e.g., core systems) to the cloud	aws Other Clued Services
	Data aggregation logic is a black box	Semantic Layer Implementation	Implement a logic layer between databases and BI tools to centralize and clarify aggregation logic for efficient management	cube Looker
	Lacking resources for BI tool development	BI Tool Development	Provide end-to-end support from UI/UX design to data integration and BI tool development	+ a b l e a u Power BI Superset
	Needs flexible, easy-to-use data apps	Data App Development	Quickly build cost-effective, widely usable data apps using frameworks like Streamlit	Streamlit
Others	External data research is time-consuming	Providing External Data List	Provide curated lists of external data sources (paid / free) across multiple sectors based on our inhouse research for deeper insights	Providing Relevant Materials MITSUI & CO.

Case in Mongolia

- GEOTRA has developed and launched “Geopro”, Mongolia’s first population movement analysis solution driven by big data.
- Geopro is a data analytics service that utilizes AI to process Mobicom’s cellular radio station network usage data, enabling detailed analysis of traffic flow, human mobility, population density, and other insights.
- This service enables optimized urban planning, infrastructure development, and population movement analysis, providing valuable solutions across multiple industries.



Disaster Prevention Use Cases

Consulting / Optimization of Emergency Radio System Placement

Utilize hazard maps and human mobility data to evaluate the optimal placement of disaster broadcast systems.

Background & Challenges

- Installing and maintaining emergency broadcast systems involves significant costs.
- We aim to evaluate whether these systems are appropriately placed in locations where they are most needed during a disaster.

Analytical Approach

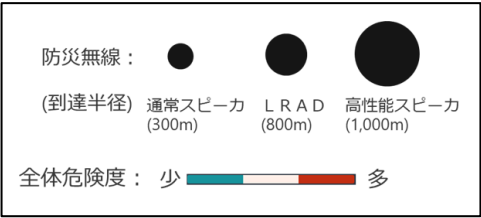
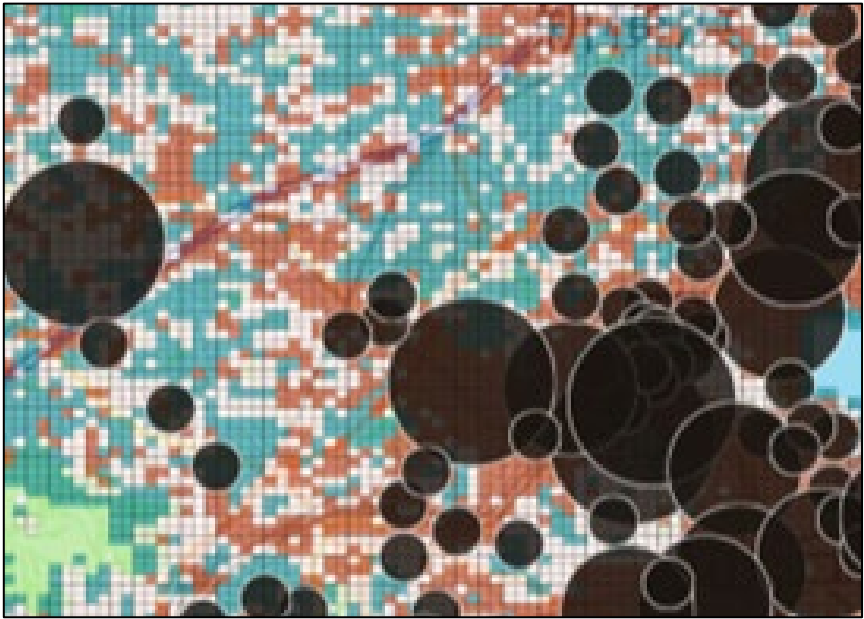
- By combining hazard maps with human mobility data—such as population density and elderly ratio—we assess the risk level of each area and overlay this with the current placement of disaster broadcast systems to verify their appropriateness.

Outcomes & Future Outlook

- By overlaying the risk level of each area with the coverage range of disaster broadcast systems, we evaluated the appropriateness of current placements and gained insights to support future relocation and supplementary measures.

Analysis Overview

Visualization of Hazard Levels and Overlay of Emergency Broadcast Coverage in Target Areas



Highlights

Evaluate the optimal placement of disaster broadcast systems based on their intended purpose, using human mobility data and hazard maps.

Understand actual evacuation behavior during past disasters to identify challenges and areas for improvement in evacuation planning.

Background & Challenges

- Utilize human mobility data and simulations to visualize evacuation routes and evaluate priorities for road restoration. This approach also supports large-scale disaster management, which is difficult to cover with AI cameras alone.

Analytical Approach

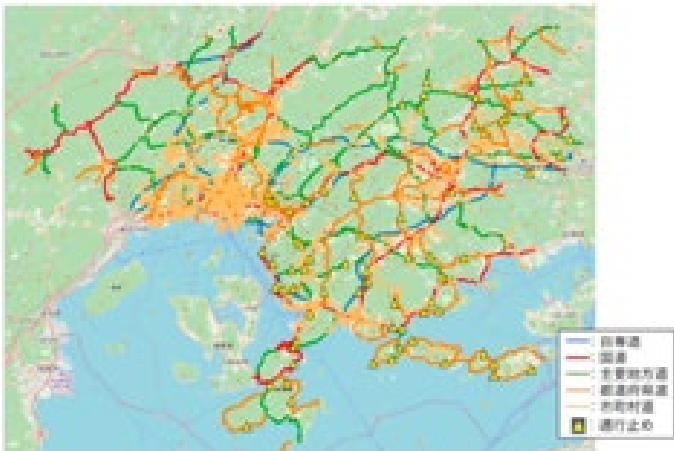
- Reconstruct disaster-time conditions using human mobility data to simulate priority routes for road restoration.
- Compare and evaluate the results against actual restoration records.

Outcomes & Future Outlook

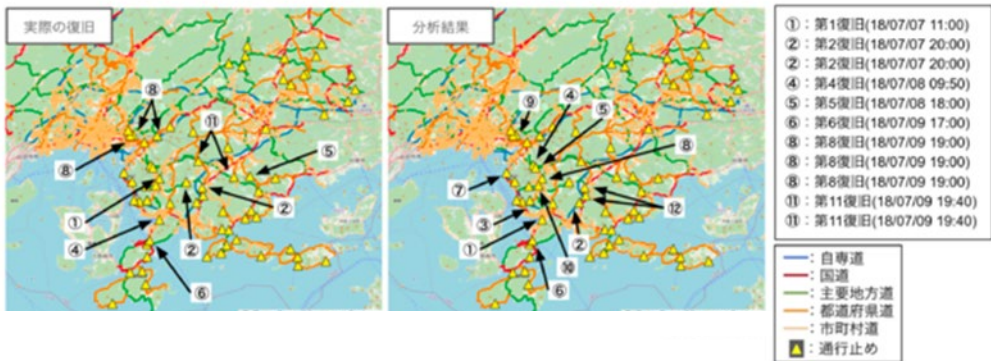
- By comparing actual responses with the ideal sequence through road clearance simulations, valuable insights were gained to enhance future prioritization and improve the efficiency of emergency response.

Analysis Overview

Simulating Human Movement and Traffic Conditions During Disasters



【防災】 シミュレーションSTEP1



Highlights

By comparing actual responses with the ideal clearance sequence, insights were gained to enhance prioritization strategies during disasters.

Toward walkable urban development and redevelopment, evaluate current visitor attributes and circulation patterns using human mobility data, and verify the impact of social experiments.

Background & Challenges

- Conduct human mobility simulations that account for building and road damage, as well as movement speed during earthquakes, to evaluate evacuation center capacity and evacuation completion times.

Analytical Approach

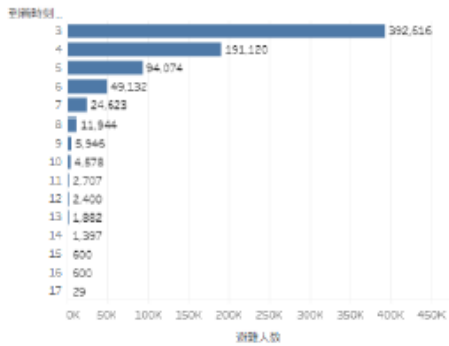
- Build a road network that reflects building collapse and road damage, extract target evacuees, and conduct human mobility simulations to recreate evacuation scenarios.

Outcomes & Future Outlook

- Simulated evacuation behavior based on damage assumptions was used to assess the adequacy of evacuation center capacity.
- Contributed to the advancement of future evacuation planning and disaster preparedness.

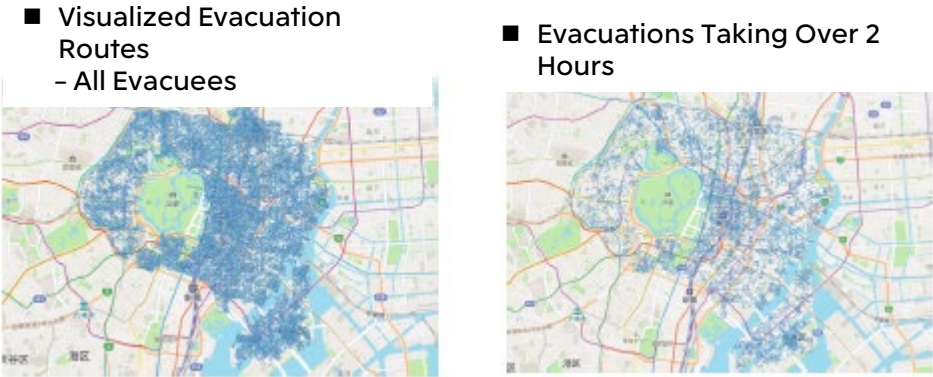
Analysis Overview

Distribution of Evacuation Completion Times



時間帯	避難率	累計避難率
3時台	50.10%	50.10%
4時台	24.39%	74.49%
5時台	12.00%	86.49%
6時台	6.27%	92.76%
7時台	3.14%	95.91%
8時台	1.52%	97.43%
9時台	0.76%	98.19%
10時台	0.58%	98.77%
11時台	0.35%	99.12%
12時台	0.31%	99.42%
13時台	0.24%	99.67%
14時台	0.18%	99.84%
15時台	0.08%	99.92%
16時台	0.08%	99.997%
17時台	0.004%	100.000%

OD Information by Evacuation Completion Time Zone



Highlights

Construct a road network based on anticipated disaster scenarios and simulate human mobility during emergencies.

360° business innovation.

